

Advanced (Habanero)

Q1. Write the equation of a line in $y = mx + b$ form that passes through the point $(2, 3)$ with slope $m = 2$.

- (A) $y = 2x + 1$
 (B) $y = 2x - 1$
 (C) $y = x + 1$
 (D) $y = x - 1$
 (E) $y = 2x + 3$

Q2. Convert the equation $y = 3x - 2$ to standard form $Ax + By = C$.

- (A) $3x - y = 2$
 (B) $-3x + y = 2$
 (C) $3x + y = 2$
 (D) $3x - y = -2$
 (E) $x + 3y = 2$

Q3. Find the slope of the line that passes through points $(1, 2)$ and $(3, 4)$.

- (A) $m = 0$
 (B) $m = 1$
 (C) $m = 2$
 (D) $m =$

$\frac{1}{2}$

(D) $m = -1$

Q4. Write the equation of a line in point-slope form $y - y_1 = m(x - x_1)$ with slope $m = 1$ and passing through $(1, 2)$.

- (A) $y - 2 = 1(x - 1)$
 (B) $y - 1 = 1(x - 2)$
 (C) $y - 1 = 2(x - 1)$
 (D) $y - 2 = 2(x - 1)$
 (E) $y = x - 1$

Q5. Graph the equation $y = -2x + 1$. What is the y -intercept?

- (A) $(0, 1)$
 (B) $(0, -1)$
 (C) $(1, 0)$
 (D) $(-1, 0)$
 (E) $(0, 2)$

Q6. Convert $2x + 3y = 7$ to slope-intercept form.

(A) $y = -$

$\frac{2}{3}x + \frac{7}{3}$

(A) $y =$

$\frac{2}{3}x + \frac{7}{3}$

(A) $y =$

$\frac{3}{2}x + \frac{7}{2}$

(A) $y = -$

$\frac{3}{2}x + \frac{7}{2}$

(A) $y =$

$\frac{2}{3}x - \frac{7}{3}$

Q7. Find the equation of a line with slope $m = -3$ and y -intercept $(0, 2)$ in slope-intercept form.

- (A) $y = -3x + 2$
 (B) $y = 3x + 2$
 (C) $y = -3x - 2$
 (D) $y = 3x - 2$
 (E) $y = 2x - 3$

Q8. Write the equation $x - 2y = 3$ in point-slope form.

(A) $y =$

$\frac{1}{2}x - \frac{3}{2}$

(A) $y = -$

$\frac{1}{2}x + \frac{3}{2}$

(A) $y =$

$\frac{1}{2}x + \frac{3}{2}$

(A) $y - 0 = -$

$\frac{1}{2}(x - 0)$

(A) $y =$

$\frac{1}{2}x - \frac{3}{2}$

Open Ended Questions (FRQ)

Q9. A water tank can hold 120 gallons of water. Due to a leak, the tank is losing water at a rate of 2 gallons per minute. If t represents time in minutes, write an equation for the amount of water W remaining in the tank after t minutes and graph it.

Q10. The cost C of renting a car for a day is given by the equation $C = 0.25m + 20$, where m is the number of miles driven. Interpret the slope and the y -intercept of this equation in the context of the problem and provide a graph.

Chef's Tips

- Emphasize the importance of labeling axes when graphing.
- Remind students to check their work by plugging in values.
- Make sure students understand the real-world applications.